Slide Set #4

Regression Models and the Classical Linear Regression Model (CLRM)

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Notes

PART I

REGRESSION MODELS

Notes

Regression modeling

Regression analysis
is a set of statistical techniques for modeling and analyzing the relationship between two types of variables:

- Y: dependent variable, response variable, outcome variable, output variable, target variable etc.
- X: independent variable, regressor, covariate, explanatory variable, predictor, feature, etc. Typically $X = (X_1, X_2, \ldots, X_K)'$ is a vector of variables.

Example
Y = expenditure of food
X = (age, household size, number of children, region, type of job, etc)$'$

Understanding the link
we assume a mechanism $X \longrightarrow Y$. We use observed data to understand it.

Notes

$Y \in \mathbb{R}$ and $X \in \mathbb{R}^K$ are random variables

Regression function f

$$Y \approx f(X)$$

models the relationship between Y and X, in the sense that f approximates Y based on X

Conditional mean regression model
a popular choice for f is

$$f(X) := \mathrm{E}[Y|X]$$

we are saying that we approximate ($\approx$) Y with its conditional expectation given the information generated by X

Notes

Why conditional mean regression models are so popular?

observe that the quality of the approximation $Y \approx f(X)$ can be measured by the quadratic risk or MSE

$$\mathrm{E}[(Y - f(X))^2]$$

Theorem (Optimal Predictor)

Under suitable regularity conditions: $\inf_f \mathrm{E}[(Y - f(X))^2] = \mathrm{E}[Y|X]$

In other words

- $f(X) = \mathrm{E}[Y|X]$ gives the best approximation to Y in terms of MSE
- If we want to guess Y based on information generated by X, $f(X) = \mathrm{E}[Y|X]$ is the best guess in terms of MSE

Notes

Proof. We need to show that for any function $f(X)$

$$\mathrm{E}[(Y - \mathrm{E}[Y|X])^2] \leq \mathrm{E}[(Y - f(X))^2]$$

$$\mathrm{E}[(Y - f(X))^2] = \mathrm{E}\left[\left(\underbrace{Y - \mathrm{E}[Y|X]}_{A} + \underbrace{\mathrm{E}[Y|X] - f(X)}_{B}\right)^2\right]$$

computing $(A + B)^2$ and using using the lineary of expectations

$$\mathrm{E}[A^2] + \mathrm{E}[B^2] + 2\,\mathrm{E}[AB] = \mathrm{E}[(Y - f(X))^2] \tag{4.1}$$

$\mathrm{E}[B^2] = \mathrm{E}[(\mathrm{E}[Y|X] - f(X))^2] \geq 0$, therefore (4.1) becomes

$$\mathrm{E}[A^2] + 2\,\mathrm{E}[AB] \;\leq\; \mathrm{E}[(Y - f(X))^2] \tag{4.2}$$

Notes

$$
\begin{aligned}
\mathrm{E}[AB] &= \mathrm{E}[\,\mathrm{E}[AB\,|\,X]\,] \quad \text{(law of iterated expectations)} \\
&= \mathrm{E}\,[\,(Y - \mathrm{E}[Y|X])(\mathrm{E}[Y|X] - f(X))\;|\;X] \\
&= (\mathrm{E}[Y|X] - f(X))\;\mathrm{E}\,[\,(Y - \mathrm{E}[Y|X])\;|\;X] \quad \text{(pull out)} \\
&= (\mathrm{E}[Y|X] - f(X))\,\{\,\mathrm{E}[Y\,|\,X] - \mathrm{E}[\mathrm{E}[Y\,|\,X]\,|\,X]\} \\
&= (\mathrm{E}[Y|X] - f(X)) \times 0
\end{aligned}
$$

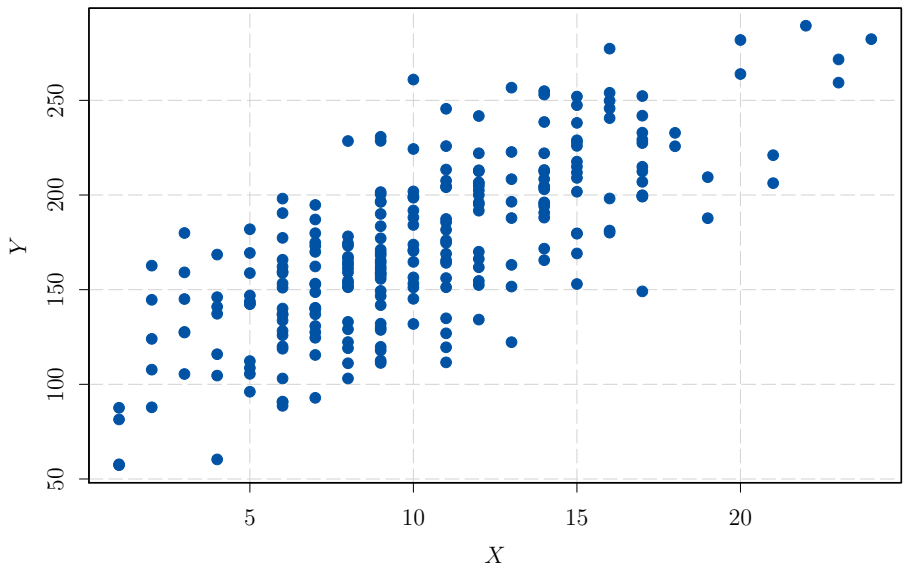
Therefore, (4.2) becomes

$$\mathrm{E}[A^2] \;\leq\; \mathrm{E}[(Y - f(X))^2]$$

with $\mathrm{E}[A^2] = \mathrm{E}[(Y - \mathrm{E}[Y\,|\,X])^2]$, which proves the result ■

Notes

Yes, but how $f(X) = \mathrm{E}[Y|X]$ looks like? Let's go back to a previous example:



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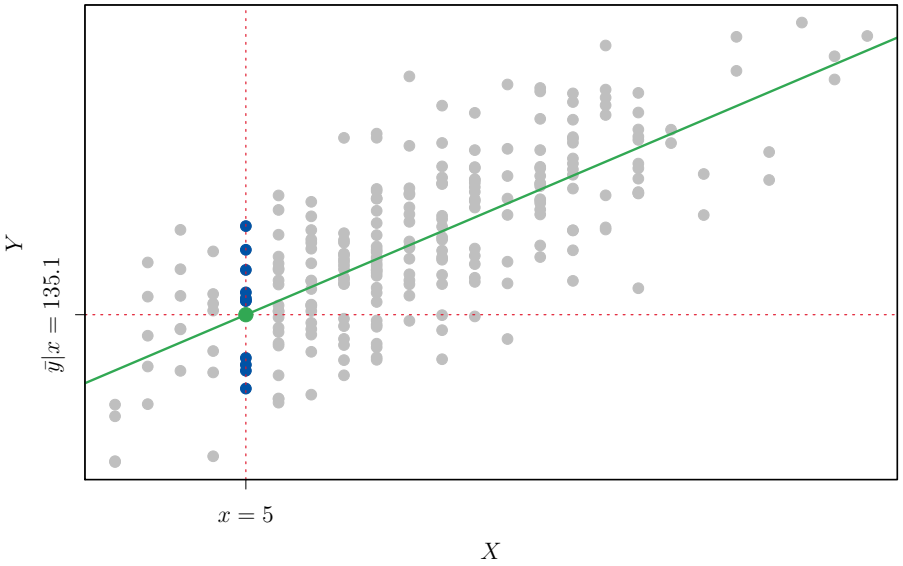

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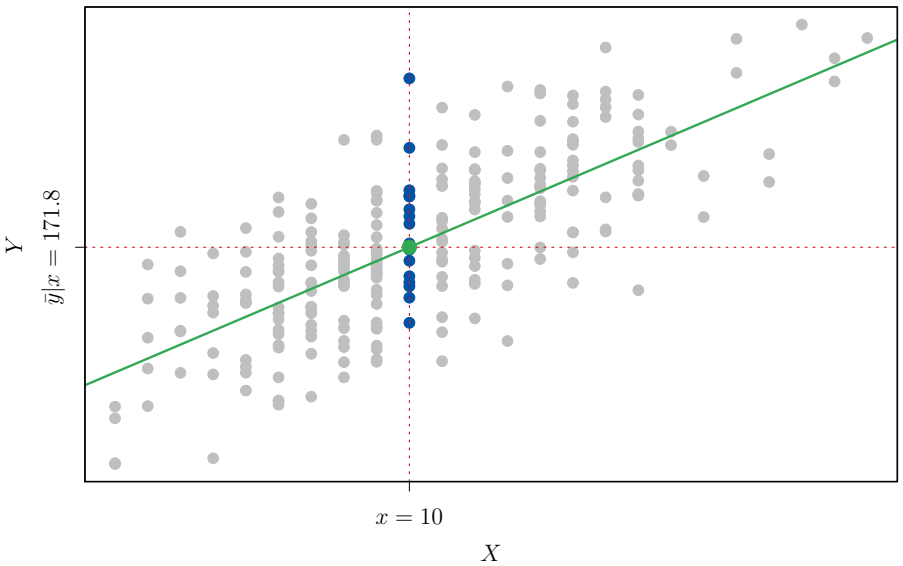

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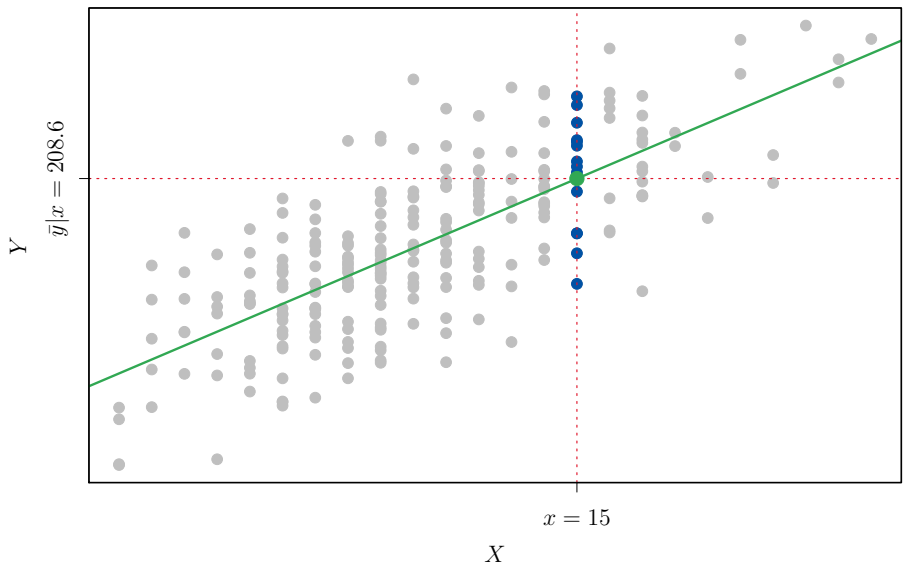

Notes

Linear (conditional mean) regression models

Linear regression function

$$f(X) \;=\; \mathrm{E}[Y \,|\, X] \;=\; \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_K X_K$$

Parameter vector $\boldsymbol{\beta}$

$\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_K)'$ describes the link between X and $\mathrm{E}[Y|X]$

Sampling

but in reality we cannot observe $\mathrm{E}[Y \,|\, X]$. For each sample unit $i = 1, 2, \ldots, n$ we observe

$$(Y_i, X_{i1}, X_{i2}, \ldots, X_{iK})$$

Notes

Reconciling the population with sampling

we add a random fluctuation, called error term, around the conditional mean

$$Y_i = \mathrm{E}[Y_i \,|\, X] + \text{error}_i$$

Average behavior of the error

taking $\mathrm{E}[\cdot \,|\, X]$ on both sides

$$\begin{aligned} \mathrm{E}[Y_i \,|\, X] &= \mathrm{E}[\mathrm{E}[Y_i \,|\, X] \,|\, X] + \mathrm{E}[\text{error}_i \,|\, X] \\ \mathrm{E}[Y_i \,|\, X] &= \mathrm{E}[Y_i \,|\, X] + \mathrm{E}[\text{error}_i \,|\, X] \end{aligned}$$

Therefore, the error term needs to be such that $\mathrm{E}[\text{error}_i \,|\, X] = 0$

Object of this course

$$Y_i = \underbrace{\beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_K X_K}_{\mathrm{E}[Y_i \,|\, X]} + \underbrace{\varepsilon_i}_{\text{error}_i}$$

Notes

PART II

Using linear regression models

Notes

Partial or marginal effects

We want to measure of the effect on $\mathrm{E}[Y \mid X]$ by a change in X_j holding all the other regressors constant (waee = "*with all else equal*").

Assuming differentiability of $f(\cdot)$, the *partial/marginal effect of a change* ΔX_j is given by

$$\Delta\, \mathrm{E}[Y \mid X] = \frac{\partial\, \mathrm{E}[Y \mid X]}{\partial X_j} \Delta X_j$$

hoding fixed $X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_K$ (*waee*)

Computing marginal/partial effects make sense only when the model has a *causal interpretation* (see later).

Notes

Continuous X_j

$$\frac{\partial\, \mathrm{E}[Y \mid X]}{\partial X_j} = \beta_j$$

Therefore, the unknown parameter β_j coincides with *partial effect* of an *unit change* in X_j *waee*.

Discrete X_j

Suppose $X_j \in \{a, b, c\}$. The partial effect when X_k changes from level a to b is

$$\Delta\, \mathrm{E}[Y \mid X] \;=\; \mathrm{E}[Y \mid X_k = b] \;-\; \mathrm{E}[Y \mid X_k = a] \qquad \text{waee}$$

these marginal effects are conveniently computed introducing dummy variables (see later)

Notes

A sample of workers went through a training program, we observe

- $\texttt{Wage}$ = measured in $10^3\$\times$USD/month
- $\texttt{Train}$ = hours spent in a training program
- $\texttt{Education}$ = time in years spent in school programs

Consider the linear model

$$\begin{aligned}\texttt{Wage} &= \beta_1 + \beta_2\texttt{Train} + \beta_3\texttt{Education} + \ldots + \varepsilon \\ &= 0.25 + 0.005(\texttt{Train}) + 0.01(\texttt{Education}) + \ldots + \varepsilon\end{aligned} \tag{4.3}$$

- What's the marginal change in expected wages when a worker add 1 unit (1h) of training ? *Answer:* $\beta_2 = 0.005$, corresponding to $5\$$/month (*waee*)
- What's the expected change in wage if a workers decides to study 3 more years? *Answer:* $\beta_3\Delta\texttt{Education} = 0.01 \times 3 = 0.03$, corresponding to $30\$$/month (*waee*)
- We want to test the null hypothesis that the training program doesn't affect wages. *Answer:* perform an hypothesis test: H_0: $\beta_2 = 0$ vs H_1: $\beta_2 \neq 0$

Notes

Dummy variables

A *dummy variable* is usually binary $\{0, 1\}$ categorical variable used identify groups in the data, or to identify *states* of the units.

A model with a dummy variable

$$Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i \tag{4.4}$$

where

$$X_i := \begin{cases} 0 & \text{if } i \in \text{Group 1} \\ 1 & \text{if } i \in \text{Group 2} \end{cases}$$

(Recall: we want $\mathrm{E}[\varepsilon_i \,|\, X] = 0$ for all $i = 1, 2, \ldots$)

Marginal effect

what's $\Delta\,\mathrm{E}[Y|X]$ when i moves from group 1 to group 2?

$$\begin{aligned}\Delta\,\mathrm{E}[Y|X] &= \mathrm{E}[Y_i \,|\, X_i = 1] - \mathrm{E}[Y_i \,|\, X_i = 0] \\ &= \beta_1 + \beta_2 \cdot 0 - (\beta_1 + \beta_2 \cdot 1) \\ &= \beta_2\end{aligned}$$

Notes

Baseline

$X = 0$ is called *baseline/reference*. Therefore, β_2 measures the expected difference in Y wrt to the baseline

Modeling mean difference/gap

Suppose Y=wage, and $X = \{0, 1\}$={Female, Male}, the previous is equivalent to

$$\begin{cases} \text{Female:} & Y_i = \beta_1 + \varepsilon_i \\ \text{Male:} & Y_i = \beta_1 + \beta_2 + \varepsilon_i \end{cases}$$

assuming $\mathrm{E}[\varepsilon_i] = 0$, $\mathrm{E}[Y|\text{Female}] = \beta_2$, and $\mathrm{E}[Y|\text{Male}] = (\beta_1 + \beta_2)$, and β_2 measures the gap in expected wages.

Polytomous variable

sometimes we have more than 2 categories. These can be encoded with 0-1-dummies. If we want to encode $G > 2$ groups we need $G - 1$ dummy variables (see homeworks).

Notes

Interaction terms

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \ldots + \varepsilon$$

adding an *interaction term* between X_2 and X_3

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_2 X_3 + \ldots + \text{error}$$

Marginal effects wrt X_2 and X_3

$$\frac{\partial\, \mathrm{E}[Y \mid \boldsymbol{X}]}{\partial X_2} = \beta_2 + \beta_4 X_3 \qquad \frac{\partial\, \mathrm{E}[Y \mid \boldsymbol{X}]}{\partial X_3} = \beta_3 + \beta_4 X_2$$

Observe: each marginal effect depends linearly on the level of the parent regressor in the interaction. What happens if $\beta_4 = 0$?

Notes

Example 1

reconsider model (4.3) and add an additional regressor equal to the interaction (Train×Education)

$$\texttt{Wage} = \beta_1 + \beta_2 \texttt{Train} + \beta_3 \texttt{Education} + \beta_4 (\texttt{Train} \times \texttt{Education}) + \ldots + \varepsilon \quad (4.5)$$

Note that

$$\frac{\partial \, \mathrm{E}[\texttt{Wage} \mid \boldsymbol{X}]}{\partial \texttt{Train}} = \beta_2 + \beta_4 \texttt{Education}$$

Suppose we want to test whether the marginal effect of the training program (Train) is proportional to the level of education (Education).

Answer: the problem is equivalent to test the null hypothesis $\mathrm{H}_0 : \beta_4 = 0$ vs $\mathrm{H}_1 : \beta_4 \neq 0$

Notes

Example 2

we want to model Income, Age and education. We have information on the highest level of education attained = {BA, MSc, PhD}.

We can transform the 3-levels categorical variable into 2 dummy variables $\{M, P\}$

Max school degree	Dummy	
	M_i	P_i
i has BA	0	0
i has MSc	1	0
i has PhD	0	1

The case "i has BA" is called *baseline*-group or *control*-group, less common is the terminology *reference*-group

Notes

First, we want to model that each degree provides an additional increase in income

$$\texttt{Income}_i = \beta_1 + \beta_2 \texttt{Age}_i + \beta_3 \texttt{M}_i + \beta_4 \texttt{P}_i + \varepsilon_i \tag{4.6}$$

The coefficient of a *dummy-coded* level expresses the variation of $\mathrm{E}[Y \mid \boldsymbol{X}]$ wrt the *baseline* level (w.a.e.e.)

- β_3: measures $\Delta\,\mathrm{E}[\texttt{Income}|\boldsymbol{X}]$ when an individual goes from BA to an MSc (waee)
- β_4: measures $\Delta\,\mathrm{E}[\texttt{Income}|\boldsymbol{X}]$ when an individual goes from BA to an PhD (waee)

Observe

⚠ Model (4.6) assumes that your school degree plays the same role for the expected income whatever it's your age.

Notes

But, if you have a PhD and you are 27 this has an expected effect on your income that is different wrt to somebody else that is a 55 with a PhD.

Let's add some interactions

$$\texttt{Income}_i = \beta_1 + \beta_2 \texttt{Age}_i + \beta_3 \texttt{M}_i + \beta_4 \texttt{P}_i + \beta_5 \texttt{Age}_i \times \texttt{M}_i + \beta_6 \texttt{Age}_i \times \texttt{P}_i + \varepsilon_i \tag{4.7}$$

Now the marginal impact of aging depends on the school degree:

$$\frac{\partial\,\mathrm{E}[\texttt{Income} \mid \boldsymbol{X}\ \&\ \texttt{M} = 0\ \&\ \texttt{P} = 0]}{\partial \texttt{Age}} = \beta_2$$

$$\frac{\partial\,\mathrm{E}[\texttt{Income} \mid \boldsymbol{X}\ \&\ \texttt{M} = 1]}{\partial \texttt{Age}} = \beta_2 + \beta_5$$

$$\frac{\partial\,\mathrm{E}[\texttt{Income} \mid \boldsymbol{X}\ \&\ \texttt{P} = 1]}{\partial \texttt{Age}} = \beta_2 + \beta_6$$

Notes

Log-linear models

(aka constant elasticity models)

$$Y = e^{\beta_1} X_2^{\beta_2} X_3^{\beta_3} e^{\varepsilon} \tag{4.8}$$

This can be *log-linearized* as follows

$$\log Y = \beta_1 + \beta_2 \log X_2 + \beta_3 \log X_3 + \varepsilon \tag{4.9}$$

Observe that the partial effect computed on (4.9) corresponds to the elasiticity of the function (4.8):

$$\frac{\partial \, \mathrm{E}[\log Y \,|\, \boldsymbol{X}]}{\partial \log X_j} = \beta_j = \frac{\partial Y}{\partial X_j} \frac{X_j}{Y}$$

In log-linear models the coefficients are elasticities. For this reason the log-linear model is also known as the "*constant elasticity model*".

Notes

Semilog-linear models

(models for relative changes)

$$Y = e^{\beta_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots + \varepsilon}$$

by taking logs

$$\log Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots + \varepsilon$$

Observe

$$\Delta \, \mathrm{E}[\log Y] = \frac{\partial \, \mathrm{E}[\log Y \,|\, \boldsymbol{X}]}{\partial X_j} \Delta X_j = \beta_j \Delta X_j$$

Interpreting coefficients in terms of rate change of Y

set $\Delta X_j = (X_j^1 - X_j^0)$ and $R_Y = (Y^1 - Y^0)/Y^0$. It can be shown that

$$e^{\beta_j \Delta X_j} - 1 \quad \leq \quad \mathrm{E}[R_Y \,|\, \boldsymbol{X}]$$

⚠ quite often I've seen people arguing that $R_Y = e^{\beta_j \Delta X_j} - 1$, but this would be right if we didn't have ε in the model (deterministic model)

Notes

PART III

CLASSICAL LINEAR REGRESSION MODEL (CLRM)

Notes

Classical linear regression model (CLRM)

Assumption (A1): linearity

For all $i = 1, 2, ..., n$ observed data are generated by the following linear model

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + ... + \beta_K x_{iK} + \varepsilon_i \tag{4.10}$$
$$= \boldsymbol{x}_i' \boldsymbol{\beta} + \varepsilon_i \tag{4.11}$$

where $\boldsymbol{\beta} \in \mathbb{R}^K$ is a vector of coefficients. In matrix form

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{4.12}$$

Remark
with x_2=log(consumption), the model is still linear wrt to log(consumption).

Notes

Remark
often a constant/intercept term is introduced in the model

$$y_i \;=\; \beta_1 + \beta_2 x_{i2} + ... + \beta_K x_{iK} + \varepsilon_i \quad \text{for } i = 1, 2, ..., n$$

In this case we consider conventionally (4.10) as if $x_{i1} = 1$ for all sample units $i = 1, 2, \ldots, n$.

Recall
if we include a constant/intercept term the first column of the data matrix $\boldsymbol{X}$ is the unit column vector

$$\boldsymbol{X} = \begin{pmatrix} 1 & x_{12} & \ldots x_{1K} \\ 1 & x_{22} & \ldots x_{2K} \\ \vdots & \vdots & \ddots \vdots \\ 1 & x_{n2} & \ldots x_{nK} \end{pmatrix}$$

Notes

Assumption (A2): strict exogeneity

$\mathrm{E}[\varepsilon_i \,|\, \boldsymbol{X}] = 0$ for all $i = 1, 2, \ldots, n$, or $\mathrm{E}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] = \boldsymbol{0}$

(A2) is a crucial assumption which can be understood as follow

Proposition (strict exogeneity means orthogonality)

Under assumption (A1) and (A2)

1. $\mathrm{E}[\varepsilon_i] = 0$ for all $i = 1, 2, \ldots, n$.
2. $\mathrm{E}[x_{jk}\varepsilon_i] = 0$, for all $i, j = 1, 2, \ldots, n$ and $k = 1, 2, \ldots, K$
3. $\mathrm{Cov}[x_{jk}, \varepsilon_i] = 0$, for all $i, j = 1, 2, \ldots, n$ and $k = 1, 2, \ldots, K$

Notes

Proof.

1. It follows from the law of iterated expectations:

$$\mathrm{E}[\varepsilon_i] = \mathrm{E}[\mathrm{E}[\varepsilon_i \,|\, \boldsymbol{X}]] = \mathrm{E}[0] = 0$$

2. By the law of iterated expectations write

$$\mathrm{E}[x_{ji}\varepsilon_i] = \mathrm{E}[\mathrm{E}[x_{jk}\varepsilon_i \,|\, x_{jk}]]$$

And by the linearity of the conditional expectation ("*pull out what's known*"), write

$$\mathrm{E}[\mathrm{E}[x_{jk}\varepsilon_i \,|\, x_{jk}]] = \mathrm{E}[x_{jk}\,\mathrm{E}[\varepsilon_i \,|\, x_{jk}]]$$

But A2 says $\mathrm{E}[\varepsilon_i \,|\, x_{jk}] = 0$, which proves the desired result.

3. This follows from previous results:

$$\mathrm{Cov}[x_{jk}, \varepsilon_i] = \mathrm{E}[x_{jk}\varepsilon_i] - \mathrm{E}[x_{jk}]\,\mathrm{E}[\varepsilon_i] = \mathrm{E}[x_{jk}\varepsilon_i] = 0$$

Notes

Remark

(A2) imposes a strong form of orthogonality: it implies that the error term of the i-th subject is uncorrelated to all possible regressors' measurements.

Suppose i =time, consider the consumption model

$$\text{Consumption}_i = \beta_1 + \beta_2\text{Wage}_i + \beta_3\text{PiggyBank}_i + \beta_4\text{LotteryWins}_i + \varepsilon_i$$

- recall: the regression function describes *the average mechanism* acting between Y and the Xs
- imagine that at time i you win a huge amount of money! Is it reasonable to assume that ε_i won't be correlated with PiggyBank_{i+1} and Wage_{i+1}?

⚠ with time series sampling strict exogeneity implies that the error term is orthogonal to the past, current, and future regressors.... This is science-fiction!

Notes

Assumption (A3): absence of multicollinearity

The $(n \times K)$ design matrix $\boldsymbol{X}$ has rank K with probability 1.

This assumption implies that $\boldsymbol{X}$ has full column rank, which means that the columns of $\boldsymbol{X}$ are linearly independent

A3 also requires that $n \geq K$.

Essentially A3 is a technical assumption that guarantees that most computations can be performed... see this later.

Notes

Assumption A4: spherical error variance

For all $i, j = 1, 2, \ldots, n$ and $i \neq j$

1. $\mathrm{E}[\varepsilon_i^2 \mid \boldsymbol{X}] = \sigma^2 > 0$ (homoskedasticity)
2. $\mathrm{E}[\varepsilon_i \varepsilon_j \mid \boldsymbol{X}] = 0$ (units are uncorrelated)

A random variable is said to have a spherical distribution if its covariance matrix is a scalar multiple of the identity matrix.

Under (A4) and (A3) we obtain

$$\mathrm{Var}[\varepsilon_i \mid \boldsymbol{X}] = \mathrm{E}[\varepsilon_i^2 \mid \boldsymbol{X}] - \mathrm{E}[\varepsilon_i \mid \boldsymbol{X}]^2 = \mathrm{E}[\varepsilon_i^2 \mid \boldsymbol{X}] = \sigma^2$$

and

$$\mathrm{Cov}[\varepsilon_i, \varepsilon_j \mid \boldsymbol{X}] = \mathrm{E}[\varepsilon_i \varepsilon_j \mid \boldsymbol{X}] - \mathrm{E}[\varepsilon_i \mid \boldsymbol{X}] E[\varepsilon_j \mid \boldsymbol{X}] = \mathrm{E}[\varepsilon_i \varepsilon_j \mid \boldsymbol{X}] = 0$$

Therefore, $\boldsymbol{\varepsilon}$ has a *spherical covariance matrix* (conditional on $\boldsymbol{X}$)

$$\mathrm{Var}[\boldsymbol{\varepsilon} \mid \boldsymbol{X}] = \mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \boldsymbol{X}] = \sigma^2 \boldsymbol{I}_n$$

Notes